MIL-E-1/215A
20 February 1970
SUPERSEDING
MIL-E-1/215
20 May 1953

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, TRANSMITTING

TYPE 832A

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Twin beam, RF amplifier

Outline, Base, Envelope		See figure 1
Cathode	---	Coated, unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	Top leads
Element	---	h	2g1	g2	k	hct	1g1	h	1a 2a

ABSOLUTE-MAXIMUM RATINGS:

Parameter: / Unit:	Ef V	Eb Vdc	Ec1 Vdc	Ec2 Vdc	(A) Ehk v	Ib mAdc	Ic1 mAdc	Pp W
C Telep:								
Maximum:	6.9 or 13.9	600	-175	250	100	75	6	10
Minimum:	5.7 or 11.3	---	---	---	---	---	---	---
C Teleg:								
Maximum:	6.9 or 13.9	750	-175	250	100	90	6	15
Minimum:	5.7 or 11.3	---	---	---	---	---	---	---
TEST CONDITIONS:	12.6	250	-35	250	---	---	---	---

ABSOLUTE-MAXIMUM RATINGS:

Parameter: / Unit:	Pi W	Pg2i W	Modulation ---	T(bulb) °C	F1 MHz	Alt ft
C Telep:						
Maximum:	22	3.4	Anode	200	200	10,000
Minimum:	---	---	---	---	---	---
C Teleg:						
Maximum:	36	5	---	200	200	10,000
Minimum:	---	---	---	---	---	---
TEST CONDITIONS:	---	---	---	---	---	---

GENERAL:

Qualification - Required

(A) denotes changes

Source: https://assist.dla.mil -- Downloaded: 202640-15T13:05Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1266	Total grid current	Eb = 400 Vdc; Ec2 = 250 Vdc; Ec1/Ib = 19 mAdc (see notes 1 and 4)	1.0	II	Ic1	0	-2	µAdc
(A) 1236	Power oscillation	Eb = 400 Vdc; Ec2 = 250 Vdc (max); Ec2/Ib = 90 mAdc; Ic1 = 2 to 6 mAdc; Rg = 8,000 to 18,000 ohms; Ef = 11.0 V; F = 200 MHz (see notes 3 and 4)	1.0	II	Po Ic2	14 0	--- 11.0	W mAdc
1256	Electrode current (anode)	Ec2 = 135 Vdc; Ec1 = -10 Vdc (see note 1)	1.0	II	Ib	18	42	mAdc
1256	Electrode current (screen)	Ec2 = 135 Vdc; Ec1 = -10 Vdc (see note 1)	1.0	II	Ic2	0.1	5.5	mAdc
1336	Heater-cathode leakage		1.0	II	Ihk	---	100	µAdc
1231	Emission	Eb = Ec1 = Ec2 = 25 Vdc (see note 1)	1.0	II	Is	80	---	mAdc
	Quality conformance inspection, part 2							
1301	Filament current	Test each unit separately	---	---	If	0.76	0.84	A
1331	Direct-interelectrode capacitance	See note 2	---	---	Cgp Cin Cout	--- 6.2 2.8	0.07 9.4 4.8	pF pF pF
1031	Low-frequency vibration	Ec1/Ib = 10 mAdc; Rp = 2,000 ohms (see note 1)	---	---	Ep	---	500	mVac
1036	Bump	Angle = 20°	---	---	---	---	---	---
(A) 1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
---	Life test	Group C; Eb = 750 Vdc; Ec2 = 250 Vdc (max); Ec2/Ib = 50 mAdc; Ic1 = 2 to 6 mAdc; Rg = 10,000 ohms; F = 200 MHz (see note 3)	---	---	---	---	---	---
---	Life-test end point (500 hours)	Power oscillation; Ef = 11 V	---	---	Po	10	---	W

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:05Z
Check the source to verify that this is the current version before use.

NOTES:

1. Test each unit separately. Control grid of unit not under test shall be connected to minus 100 volts.
2. Test each unit separately. Tie unit not under test to ground. Cgp shall be measured with a shield 3/4 inch (19.05 mm) high and 2-3/8 inches (60.33 mm) ID.
3. Use push-pull self-excited oscillating circuit. Po shall be useful power output.
4. This test to be performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL
Navy -
Air Force - 11, 70, 85
DSA - ES

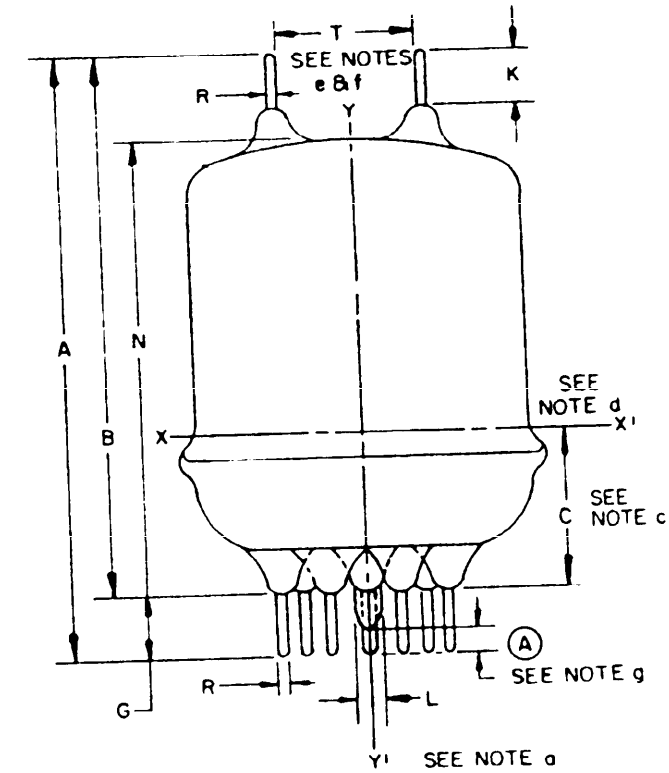
User activities:
Army - WC, AV, ME
Navy - AS, OS, MC, CG SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2401-12)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:05Z
Check the source to verify that this is the current version before use.



Ltr	Minimum	Maximum
Dimensions in inches with metric equivalents (mm) in parentheses		
Quality conformance inspection, part 1		
Ⓐ C	see note c	
F	see note b	
T	see notes e and f	
U	50°55'	51°5'
V	51°55'	52°5'
W	25°55'	26°5'
Quality conformance inspection, part 2		
A	3.063(77.80)	3.313 (84.15)
E		1.156 (29.36)
Quality conformance inspection, part 3 (periodic check)		
B	2.625 (66.68)	2.875 (73.03)
G	.375 (9.53)	.500 (12.70)
K	.375 (9.53)	.500 (12.70)
L		.375 (9.53)
N	2.000 (50.80)	2.375 (60.33)
R	.052 (1.32)	.060 (1.52)
S	.122 (3.10)	.128 (3.25)

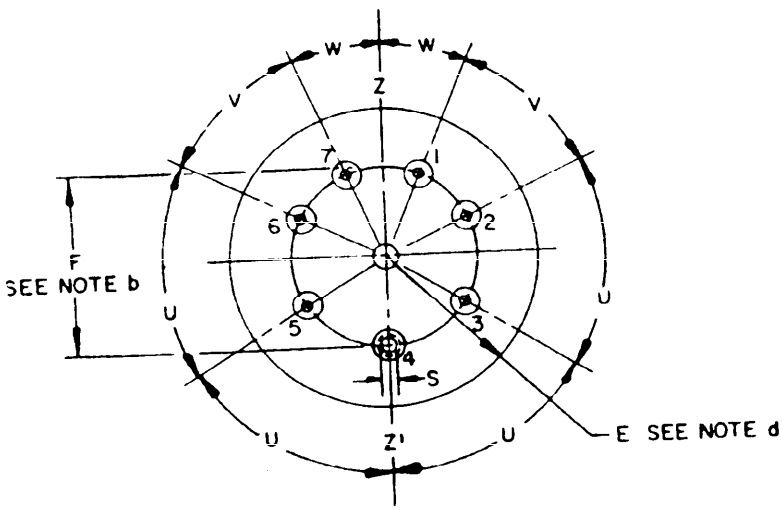


FIGURE 1. Outline drawing of electron tube type 832A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:05Z
Check the source to verify that this is the current version before use.

NOTES:

a. The axis YY' is defined as the axis of the base pin gage described in b.

b. The tube base shall be capable of entering to a distance of .375 (9.53 mm) a flat-plate gage having 6 holes .0800 (2.03 mm) ±.0005 (.01 mm) and 1 hole .1450 (3.68 mm) ±.0005 (.01 mm) arranged on a 1.000 (25.40 mm) ±.0005 (.01 mm) circle at specified angles on the outline. A .500 (12.70 mm) ±.01 (.25 mm) hole at the center of the pin circle is also required. The axis YY' is defined by the center of this hole.

c. Dimension 'C' is measured by inserting the tube in the base-pin gage described in b. and then lowering a gage plate having a hole 2.063 (52.40 mm) +.003 (.08 mm) −.000 (.00 mm) in diameter until the plate rests on the seal flange at position XX'. The center line of the hole shall be coincident with the axis YY' within .150 (3.81 mm). With the gage plate parallel to the top surface of the base pin gage, the dimension 'C' is measured between the bottom surface of the gage plate and the top surface of the base pin gage. This distance shall be .844 (21.44 mm) min. and 1.219 (30.96 mm) max.

d. Minimum diameter of tube-seal flange will be such that a gage having ID of 2.125 (53.98 mm) +.003 (.08 mm) −.000 (.00 mm) and thickness of .125 (3.18 mm) ±.010 (.25 mm) will not pass the flange when tried at any angle.

(A) e. The plate leads shall be capable of entering a flat plate gage of .375 (9.53 mm) min. thickness having 2 holes .200 (5.08 mm) ±.0005 (.01 mm) in diameter arranged .424 (10.77 mm) ±.001 (.03 mm) from a point coincident with the axis YY'. The axes of the holes shall be parallel to YY' and the plane of these axes shall be 90° ±5' to planes YY' and ZZ'.

f. The plate leads shall be capable of entering a flat plate gage of .375 (9.53 mm) min. thickness having 2 holes .120 (3.05 mm) ±.0005 (.01 mm) in diameter arranged .848 ±.001 (21.54 ±.03 mm) (center to center).

(A) g. The tip of the tube-seal shall not extend beyond the tips of the base pins.

(A) h. The plane of the electrodes of each unit shall be parallel to a plane (ZZ') passing through axis of tube and pin No. 4.

FIGURE 1. Outline drawing of electron tube type 832A - Continued.

☆ U. S. GOVERNMENT PRINTING OFFICE: 1970-393-095/5-4567

Source: https://assist.dla.mil -- Downloaded: 20260-15T13:05Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:05Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 202610-15T13:05Z
Check the source to verify that this is the current version before use.

FOLD

)EPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:05Z
Check the source to verify that this is the current version before use.